

Reviewer Pack — Noncommutative Rank Bounded by Disjointness Communication Co...

Entry 8ac0ba31a882 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Noncommutative Rank Bounded by Disjointness Communication Complexity

Entry ID: 8ac0ba31a882 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-11 12:25:58 UTC

1. Conjecture

Field A (mathematical branch): Noncommutative Geometry **Field B** (complexity object): Communication Complexity of Disjointness

Statement:

For any $n \times n$ Boolean matrix M arising from the disjointness problem, its noncommutative rank (minimal number of rank-1 matrices needed to express M as a sum in the matrix algebra over the ring of 2×2 matrices) satisfies $\text{rank_noncommutative}(M) \leq 2^{\{n/2\}} / (\text{disjointness_commsize}(M))$

Rationale (proposer's reasoning):

Noncommutative rank captures algebraic structure beyond classical linear algebra, potentially exposing hidden symmetries in communication matrices. Disjointness has optimal $\Omega(n)$ lower bounds, suggesting an inverse relationship between algebraic complexity and communication cost.

Taxonomy category: COMM_COMPLEXITY (status at proposal time: alive)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): ab5ac848c0402f58

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

default: $\geq 80\%$ seeds must support, no counterexample

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.00	UNCERTAIN	UNCERTAIN

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
NATURAL_PROOFS	SAFE	0.00	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.00	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.95	SAFE	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=1, elapsed=2.6s

5.1 Generated Python source

```
import random
import math
from itertools import product

def generate_disjointness_instance(n):
    A = [[random.randint(0, 1) for _ in range(n)] for _ in range(n)]
    B = [[random.randint(0, 1) for _ in range(n)] for _ in range(n)]
    M = [[A[i][j] ^ B[i][j] for j in range(n)] for i in range(n)]
    return M

def matrix_multiply(A, B):
    n = len(A)
    C = [[0 for _ in range(n)] for _ in range(n)]
    for i in range(n):
        for j in range(n):
            for k in range(n):
                C[i][j] += A[i][k] * B[k][j]
    return C

def is_rank_1(A, B):
    n = len(A)
    for i in range(n):
        for j in range(n):
            if A[i][j] != 0 and (A[i][j] * B[j][i] == 0 or A[i][j] * B[j][i] != A[0][0]):
                return False
    return True

def noncommutative_rank(M, max_k=10):
    n = len(M)
    for k in range(1, max_k + 1):
        for A_combination in product([0, 1], [0, -1], repeat=k * n):
            A = [[A_combination[i][j] for j in range(k)] for i in range(k)]
            B = [[A_combination[i + k][j] for j in range(k)] for i in range(k)]
            if matrix_multiply(A, B) == M:
                return k
    return float('inf')

def run_trial(seed: int) -> dict:
```

```

random.seed(seed)
n = 10
M = generate_disjointness_instance(n)
rank_noncommutative = noncommutative_rank(M)
bound = 2 ** (n / 2) / n
conjecture_holds = rank_noncommutative <= bound
counterexample = "" if conjecture_holds else f"Noncommutative rank {rank_noncommutative} > {bound}"
return {
    "metric_name": "noncommutative_rank",
    "metric_value": rank_noncommutative,
    "instances_tested": 1,
    "conjecture_holds": conjecture_holds,
    "counterexample": counterexample
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2**i + 3 for i in range(5, 8)]
    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_value = sum(r["metric_value"] for r in results) / len(results)
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)
    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std=0.0 support_fraction=1.0")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_value} std=0.0 support_fraction={support_fraction}")
    else:
        first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample={results[0]['counterexample']} first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

--STDERR--

Traceback (most recent call last):

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_73f8d0b0.py", line 71, in <module>

```

    result = run_trial(seed)
    ~~~~~^

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_73f8d0b0.py", line 54, in run_trial

```

    rank_noncommutative = noncommutative_rank(M)
    ~~~~~^

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_73f8d0b0.py", line 44, in noncommutative_rank

```

    A = [[A_combination[i][j] for j in range(k)] for i in range(k)]
    ~~~~~^

```

IndexError: list index out of range

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Test crashed with IndexError before producing results, preventing evaluation of conjecture validity | next: Fix the noncommutative_rank function to handle matrix dimensions properly and re-run tests

11. Audit log (LLM calls)

Total LLM calls: 8

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	qwen3:8b	0	0	111193
2	propose	ollama_remote	qwen3:8b	0	0	114643
3	preregistration	ollama_remote	qwen3:8b	0	0	24086
4	novelty	ollama_remote	qwen3:8b	0	0	20660
5	novelty	ollama_remote	qwen3:8b	0	0	14343
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	16607
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9203
8	judge	ollama_remote	qwen3:8b	0	0	15653

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 326388 ms total latency. Provider mix: {'ollama_remote': 8}

(full prompt+response transcripts available in research/audit/8ac0ba31a882.jsonl)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/8ac0ba31a882.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/8ac0ba31a882.tar.gz (if generated)